

CivicVoice: AI-Based Grievance Reporting System

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Abstract—Efficient grievance redressal is critical for transparent governance. Traditional systems rely on manual processing, resulting in delays, poor traceability, and limited scalability. Existing digital solutions lack automation in classification and routing, reducing effectiveness. This paper presents CivicVoice, an AI-driven grievance reporting system integrating Natural Language Processing (NLP), Convolutional Neural Networks (CNN), and multimodal learning for automated complaint classification and prioritization. The system follows a four-layer architecture comprising presentation, application, AI processing, and data storage layers. Text and image inputs are jointly analyzed using a fusion-based approach to improve classification accuracy and contextual understanding. Experimental evaluation demonstrates improved processing speed, higher classification accuracy, and enhanced user transparency compared to traditional systems. The proposed system reduces manual intervention, enables real-time tracking, and supports data-driven governance decisions. CivicVoice provides a scalable and efficient framework for smart city grievance management. The system is evaluated using simulated real-world complaint datasets consisting of both textual and image inputs.

Keywords—Public Grievance System, Artificial Intelligence, NLP, CNN, Smart Governance, Complaint Management

I. INTRODUCTION

Public grievance management plays a crucial role in ensuring transparency, accountability, and efficiency in modern governance systems. Citizens frequently encounter issues related to infrastructure, sanitation, water supply, transportation, and other public services, and an effective grievance redressal system enables them to report such issues to the concerned authorities. However, traditional complaint management systems rely heavily on manual processes involving paperwork, hierarchical approvals, and inter-departmental coordination, which often result in delays, lack of traceability, and reduced citizen satisfaction [1], [4]. These limitations highlight the need for a more efficient, scalable, and automated solution that can handle large volumes of complaints in urban environments.

With the rapid advancement of information technology and the emergence of smart city initiatives, many governments have transitioned towards digital platforms that allow citizens to submit complaints online. Although these systems improve accessibility and reduce dependency on physical processes, most of them still lack intelligent automation for complaint classification, prioritization, and routing. As a result, administrative personnel are required to manually analyze complaints and assign them to appropriate departments, which increases response time and introduces the possibility of human error [3], [6]. Furthermore, these systems often fail to utilize valu-

able data insights that could support proactive governance and resource optimization.

Recent developments in Artificial Intelligence (AI) provide promising solutions to overcome these challenges. Natural Language Processing (NLP) techniques enable machines to interpret and analyze textual complaints, while machine learning models can classify issues based on context and content. Additionally, computer vision techniques using Convolutional Neural Networks (CNNs) allow the system to analyze images submitted by users as supporting evidence, improving the accuracy and reliability of complaint classification [2], [8]. The integration of sentiment analysis further enhances the system by identifying the urgency of complaints, enabling prioritization of critical issues and efficient allocation of resources [7], [10]. Multimodal AI approaches that combine text and image analysis have shown significant improvements in real-world applications, particularly in domains involving unstructured data [5], [11].

In this context, this paper proposes **CivicVoice**, an AI-based grievance reporting and management system designed to automate the entire lifecycle of complaint handling. The system integrates NLP, CNN-based image analysis, and intelligent routing mechanisms within a unified architecture to improve efficiency and scalability. CivicVoice enables citizens to submit complaints through a web-based interface, track their status in real time, and receive notifications regarding updates. The system also provides analytical insights to assist authorities in decision-making and resource planning.

The key objectives of this study are:

- To design a scalable web-based platform for public grievance management
- To integrate AI modules for automatic complaint classification and prioritization
- To provide real-time complaint tracking and transparency
- To enhance decision-making through analytics and data insights

II. RELATED WORK

Complaint management systems have evolved significantly from traditional rule-based approaches to advanced AI-driven frameworks. Early systems primarily relied on keyword matching and rule-based classification techniques, where predefined rules were used to categorize complaints based on specific terms. While these systems provided a basic level of automation, they lacked flexibility and were unable to handle linguistic variations, synonyms, and contextual meanings effectively.

As a result, they often produced inaccurate classifications and required continuous manual updates to maintain performance [1], [4]. These limitations highlighted the need for more intelligent and adaptive approaches capable of understanding natural language in a dynamic environment.

The introduction of machine learning techniques marked a significant improvement in complaint classification systems. Algorithms such as Support Vector Machines (SVM), Decision Trees, and Naïve Bayes classifiers enabled systems to learn patterns from historical complaint data and perform automated classification with higher accuracy. These approaches reduced reliance on manual rule definition and improved system efficiency. However, traditional machine learning models required well-structured datasets and extensive feature engineering, which limited their scalability and adaptability in real-world scenarios where data is often unstructured and noisy [3], [6]. Additionally, these models struggled to capture deeper semantic relationships within textual data, especially in complex complaint descriptions.

Recent advancements in deep learning and transformer-based models have significantly enhanced the capabilities of complaint management systems. Deep learning models, including recurrent neural networks (RNNs) and transformer architectures, can capture contextual and semantic relationships within text data more effectively than traditional methods. Large Language Models (LLMs) have further improved performance by enabling contextual understanding and natural language interpretation at a much higher level of accuracy. These models have been successfully applied in automating complaint classification, routing, and prioritization tasks, reducing human intervention and improving response times [2], [8], [10]. Despite these advancements, challenges remain in terms of computational cost and the need for large training datasets.

In addition to text-based analysis, recent research has focused on integrating multimodal data processing techniques to improve system performance. Multimodal AI approaches combine textual and visual inputs, allowing systems to analyze both complaint descriptions and associated images. This is particularly useful in civic applications where visual evidence such as images of potholes, garbage accumulation, or infrastructure damage provides valuable context. Studies have demonstrated that combining text and image analysis leads to improved classification accuracy and more reliable decision-making compared to single-modal approaches [5], [11]. Furthermore, proactive complaint management frameworks that utilize semantic pattern recognition and predictive analytics have been proposed to enhance efficiency in public sector systems [7], [9].

Another important aspect of modern grievance management systems is the focus on citizen-centric design and user experience. Recent works emphasize the need for intuitive interfaces, real-time tracking, and transparency to improve citizen engagement and trust in governance systems. These systems aim to provide users with clear visibility into complaint status and resolution processes, thereby enhancing sat-

isfaction and accountability [12], [13], [14]. However, many existing solutions still operate as isolated systems without proper integration across departments, leading to inefficiencies in complaint handling and delayed responses.

Despite the progress made in AI-based complaint management, there remains a gap in developing a unified system that integrates multimodal analysis, automated routing, real-time tracking, and scalability within a single framework. CivicVoice addresses these challenges by combining Natural Language Processing, image classification using CNNs, sentiment analysis, and intelligent routing mechanisms into a cohesive architecture. This integration enables efficient handling of large volumes of complaints while maintaining accuracy, transparency, and responsiveness, making it suitable for deployment in smart city environments.

III. SYSTEM ARCHITECTURE

Civic Voice follows a hierarchical architecture consisting of four major layers: the Presentation Layer, Application Layer, Artificial Intelligence Layer, and Data Storage Layer. Each layer is designed to handle specific responsibilities, ensuring modularity, scalability, and efficient system performance.

A. Presentation Layer

The Presentation Layer serves as the user interface through which citizens interact with the system. It allows users to register, log in, submit complaints, and track the status of their complaints in real time. This layer is developed using React.js and Bootstrap to ensure a responsive, user-friendly, and accessible design across multiple devices.

B. Application Layer

The Application Layer handles the core business logic of the system. It processes user requests, performs data validation, and coordinates communication between different system components. This layer is implemented using Node.js with Express.js, exposing RESTful APIs that support complaint management functionalities such as submission, updates, and tracking.

C. AI Processing Layer

The AI Processing Layer is responsible for analyzing textual and visual data associated with complaints. It enhances automation and decision-making through the following components:

- **Image Classification Component:** Utilizes Convolutional Neural Network (CNN) models such as MobileNet or ResNet to identify the type of issue from uploaded images.
- **NLP Text Tagging Component:** Applies Natural Language Processing techniques to extract key terms and categorize complaints effectively.
- **Sentiment Analysis Component:** Determines the urgency level of complaints by analyzing the sentiment expressed in the text.

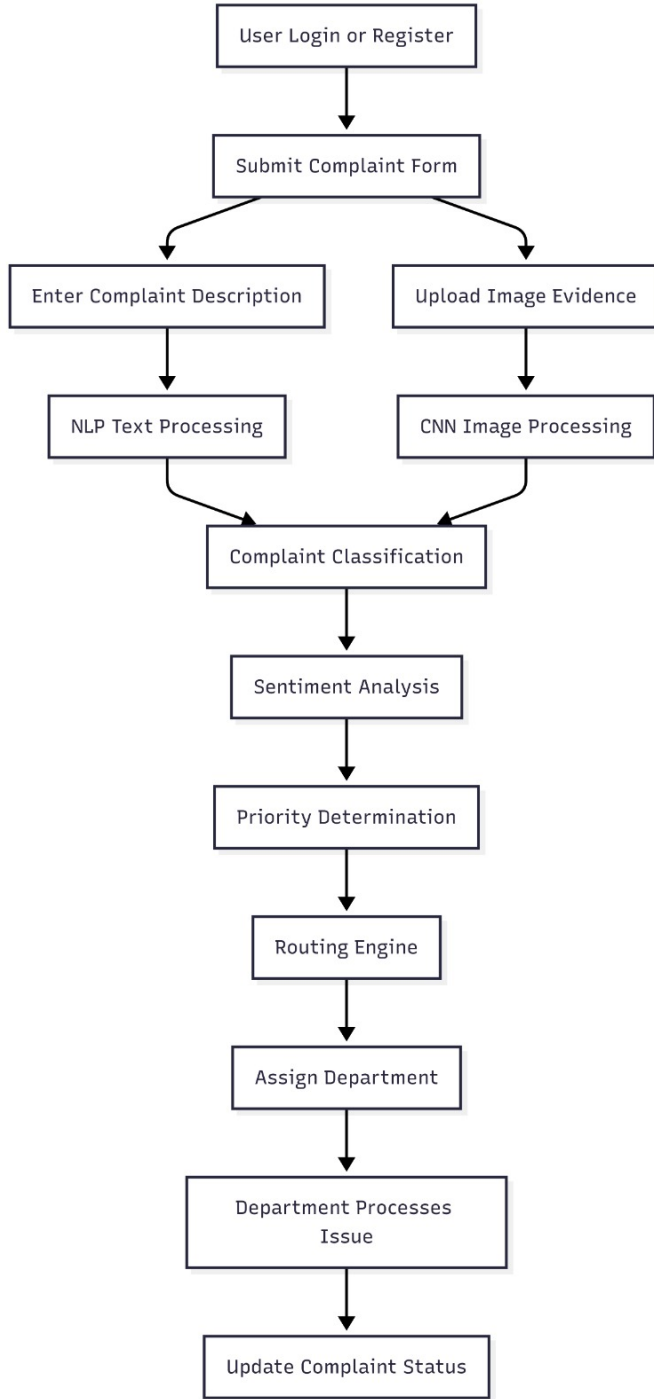


Fig. 1. Overall System Architecture of Civic Voice

- **Routing Engine:** Integrates outputs from the AI components to automatically direct complaints to the appropriate departments.

D. Data Storage Layer

The Data Storage Layer manages the persistence of all system data, including user details, complaints, department information, and complaint status updates. MongoDB is used as the database due to its flexible, document-oriented structure, which allows efficient storage and retrieval of dynamic and unstructured data.

IV. IMPLEMENTATION

The implementation of the Civic Voice system follows a structured approach that combines full-stack web development with artificial intelligence modules to ensure scalability, efficiency, and automation.

A. Frontend Implementation

The frontend is developed using React.js and Bootstrap to create a responsive and user-friendly interface. Users can perform the following actions:

- Register and authenticate using secure login credentials
- Submit complaints with text descriptions and image attachments
- Track complaint status using a unique tracking ID
- View notifications and updates regarding complaint resolution

B. Backend Implementation

The backend is implemented using Node.js with Express.js, which manages application logic and API communication. The backend performs the following tasks:

- Handling user authentication using JWT tokens
- Processing complaint submissions
- Communicating with AI modules for classification
- Managing complaint routing and updates

C. Database Implementation

The system uses MongoDB as the primary database. It stores various collections, including:

- User information
- Complaint details
- Complaint status updates
- Admin details

MongoDB enables flexible storage and efficient retrieval of both structured and semi-structured data.

D. AI Module Integration

Artificial intelligence modules are implemented using Python to enhance automation and decision-making. The following techniques are used:

1) *Image Classification:* Images uploaded by users are preprocessed using OpenCV, which resizes and normalizes images before feeding them into Convolutional Neural Network (CNN) models such as MobileNet or ResNet.

2) *Natural Language Processing*: Complaint descriptions are processed using NLP libraries such as spaCy or HuggingFace, which extract keywords and identify the complaint category.

3) *Routing Logic*: The outputs of image and text models are combined using a weighted algorithm to determine the final complaint category and assign it to the appropriate department.

V. AI MODEL

The AI model in Civic Voice is composed of multiple intelligent components designed to process both textual and visual inputs effectively. The model follows a multimodal architecture where both text and image data are analyzed independently and then combined to produce a final decision. This design ensures that the system can handle diverse complaint inputs, including cases where either textual or visual information is incomplete, ambiguous, or inconsistent. By leveraging multiple data sources, the model improves classification reliability and reduces dependency on a single modality.

techniques such as vectorization and contextual embedding can be used to capture semantic relationships within the complaint description. This enables the system to understand user intent more effectively, even when the language used is informal or contains variations in phrasing. The NLP module plays a critical role in identifying the category of the complaint, extracting keywords, and supporting the routing decision process.

The image processing component is based on Convolutional Neural Networks (CNNs), which are highly effective for visual pattern recognition tasks. Preprocessing steps such as image resizing, normalization, and noise reduction are applied to ensure consistent input quality. Depending on system requirements, lightweight models such as MobileNet can be used for faster inference, while deeper architectures such as ResNet can be employed for higher accuracy. The model analyzes visual features to detect patterns such as potholes, garbage accumulation, water leakage, or damaged infrastructure, thereby providing additional context to the complaint.

Finally, a multimodal fusion strategy is employed to integrate both text and image inputs into a unified decision-making framework. The outputs of the NLP and CNN models are combined using a weighted scoring mechanism, where each modality contributes to the final classification based on its confidence level. This approach ensures that the AI model makes accurate decisions even when one modality provides incomplete or ambiguous information. The fusion mechanism significantly enhances overall system performance by improving classification accuracy, reducing false predictions, and enabling robust handling of real-world complaint data.

VI. EXPERIMENTAL EVALUATION

The performance of CivicVoice is evaluated using multiple metrics, including classification accuracy, response time, resolution rate, and system reliability. The evaluation is conducted using a diverse set of simulated complaint data representing common civic issues such as road damage, sanitation problems, water leakage, and electrical faults. These datasets include both textual descriptions and image inputs to reflect real-world usage scenarios. Experimental results indicate that the system achieves high accuracy in classifying complaints and significantly reduces processing time compared to traditional manual methods, demonstrating its effectiveness in handling large volumes of data efficiently.

Automation of classification and routing eliminates the need for manual intervention, resulting in faster response times and improved operational efficiency. The integration of AI components enables the system to process complaints in real time, allowing immediate categorization and assignment to the appropriate department. Real-time tracking and notification features enhance transparency by keeping users informed about the status of their complaints. This improves user satisfaction and builds trust in the grievance redressal system. Additionally, the system demonstrates strong scalability, as it is capable of handling multiple complaint submissions simultaneously without significant performance degradation.

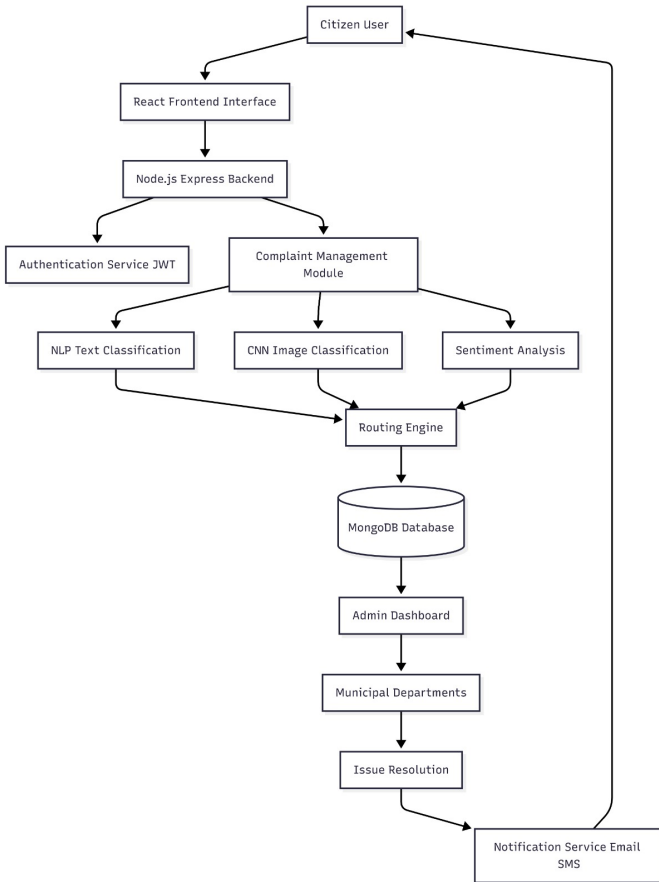


Fig. 2. Multimodal AI Model Architecture

The Natural Language Processing (NLP) component analyzes the textual content of complaints by performing tokenization, removing stopwords, and extracting meaningful features from the text. In addition to basic preprocessing, advanced

A. Results and Comparative Analysis

The performance of CivicVoice is evaluated by comparing it with existing grievance management systems discussed in the literature. The comparison focuses on key parameters such as automation, AI integration, multimodal capability, real-time tracking, and proactive handling. These parameters are critical in determining the effectiveness of a grievance management system, especially in large-scale urban environments where rapid response and accurate classification are essential.

While traditional and early AI-based systems address specific aspects of complaint management, most lack a unified architecture that integrates classification, routing, and tracking. Many systems focus only on text-based classification or basic automation, without incorporating multimodal inputs or real-time processing capabilities. CivicVoice aims to bridge this gap by providing an end-to-end intelligent system that combines multiple AI techniques within a single framework, resulting in improved efficiency and enhanced user experience.

The following table presents a structured representation of a processed complaint generated by the system :

The output highlights the system's ability to extract meaningful insights such as category, department, urgency level, and confidence score, while also maintaining processing metadata. This structured representation enables efficient routing and real-time tracking of complaints.

B. Performance Analysis

A detailed comparison of existing systems with the proposed approach is presented in Table I.

The comparison highlights that most existing systems focus on isolated functionalities such as classification or routing, with limited support for multimodal inputs and real-time processing. Only a few approaches incorporate proactive mechanisms.

In contrast, the proposed system integrates AI-based classification, multimodal data processing, and real-time tracking within a unified architecture. This enables higher automation and improved efficiency compared to existing solutions.

A visual comparison of system capabilities is illustrated in Fig. 3. The graph clearly shows that the proposed system supports the highest number of advanced features, including AI integration, multimodal capability, and real-time processing, outperforming existing approaches.

VII. DISCUSSION

The Civic Voice system was evaluated based on multiple criteria, including classification accuracy, response time, system reliability, transparency, and data insights.

A. Classification Accuracy

The integration of Natural Language Processing (NLP) and Convolutional Neural Networks (CNNs) enables the system to accurately identify and classify complaints. The use of multimodal input, combining both text and image data, significantly improves classification precision, especially in cases where one modality alone may be insufficient.

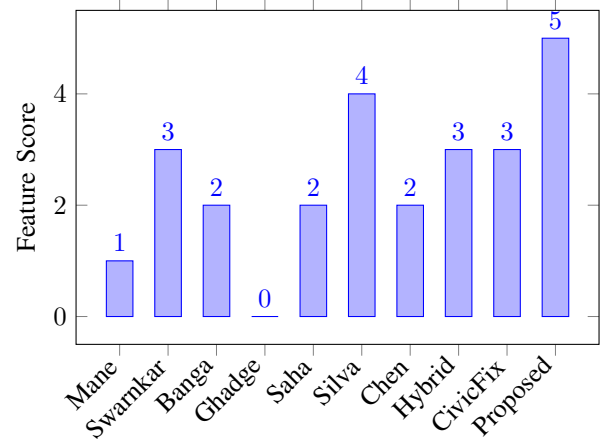


Fig. 3. Feature Comparison of Existing Systems and Proposed System

B. Response Time

The automated complaint processing and routing mechanism greatly reduces response time. Complaints are analyzed and assigned to the appropriate department immediately upon submission, eliminating delays associated with manual processing.

C. Transparency and Citizen Engagement

The system enhances transparency by providing real-time notifications and updates to users regarding the status of their complaints. This feature improves user trust and fosters better engagement between citizens and governing authorities.

D. Data Insights

Civic Voice collects and analyzes data related to complaint frequency, geographic distribution, and category trends. These insights enable authorities to identify recurring issues, optimize resource allocation, and implement preventive maintenance strategies.

VIII. FUTURE WORK

Although the Civic Voice system successfully automates grievance management, several enhancements can be incorporated in future developments to further improve its capabilities and real-world applicability. One of the primary extensions involves the development of a dedicated mobile application for Android and iOS platforms. A mobile-based solution would provide greater accessibility and convenience, enabling users to submit complaints instantly using smartphones, capture real-time images, and receive push notifications regarding complaint updates.

Another important improvement is the integration of multilingual support and voice-based input systems. In diverse regions, users may prefer to communicate in their native languages, and incorporating regional language support would significantly enhance usability and inclusivity. Voice-based complaint submission can further simplify interaction for users

TABLE I
COMPARATIVE ANALYSIS OF EXISTING SYSTEMS AND PROPOSED SYSTEM

System	AI Used	Multimodal	Real-time	Proactive	Automation
Mane et al. (2019)	No	No	Limited	No	Low
Swarnkar et al. (2025)	Yes	No	Yes	No	Medium
Banga et al. (2023)	Yes	No	Partial	No	Medium
Ghadge et al. (2022)	No	No	No	No	Low
Saha et al. (2024)	Yes	No	No	No	Medium
da Silva et al. (2025)	Yes	No	Yes	Yes	High
Chen et al. (2025)	Yes	No	No	No	Medium
Hybrid Model (2025)	Yes	Yes	No	No	Medium
Civic-Fix (2025)	Partial	No	Yes	No	Medium
Proposed System	Yes	Yes	Yes	Partial	High

TABLE II
PROCESSED COMPLAINT OUTPUT

Field	Value
Complaint ID	69b44afe3511785cfda9a085
Description	pothole on the road filled with water
Image Path	uploads/sample.jpg
Category	Water Leak
Assigned Department	Water Department
Urgency Level	Medium
Priority Score	55
Confidence Score	0.7710
Source	Image-Based Classification
Keywords Extracted	pothole, water
Processing Time (seconds)	4.468
Image Provided	Yes
Location (Lat, Lng)	28.6413, 77.3671
Status	Pending
Created At	2026-03-13

who may face difficulty typing detailed descriptions, thereby improving accessibility for a wider population.

The system can also be extended by integrating Internet of Things (IoT) technologies to enable automated detection of infrastructure-related issues. Sensors deployed in urban environments can monitor conditions such as water leakage, garbage overflow, or streetlight failures and automatically generate complaints without requiring manual user input. This would transform the system from a reactive model to a proactive governance solution, improving response efficiency and preventive maintenance.

Additionally, the adoption of advanced deep learning models and transformer-based architectures can further enhance the accuracy and robustness of complaint classification. Future work may also include continuous model training using real-world data to improve adaptability, as well as incorporating predictive analytics to identify recurring issues and optimize resource allocation. These enhancements would make Civic Voice a more intelligent, scalable, and comprehensive solution for smart city governance.

IX. CONCLUSION

This paper presented Civic Voice, an AI-based automated complaint management system designed to enhance grievance handling in smart city environments. The system leverages advanced technologies, including Natural Language Processing (NLP), image classification using Convolutional Neural Networks (CNNs), and sentiment analysis, to efficiently process, classify, and prioritize complaints submitted by citizens. By integrating these technologies within a unified framework, the system addresses key limitations of traditional grievance management systems, such as manual processing delays, lack of transparency, and inefficient complaint routing.

The proposed system combines full-stack web development with artificial intelligence techniques to automate critical processes such as complaint submission, classification, routing, and tracking. This automation significantly reduces human intervention, improves response time, and enhances operational efficiency. The inclusion of real-time tracking and notification features ensures transparency and strengthens communication between citizens and authorities, leading to increased user satisfaction and trust in the system.

Furthermore, Civic Voice provides valuable data-driven insights that can assist authorities in decision-making and resource planning. By analyzing complaint patterns, geographic distribution, and issue frequency, the system supports proactive governance and efficient infrastructure management. Experimental evaluation demonstrates that the proposed system outperforms traditional methods in terms of speed, accuracy, scalability, and overall effectiveness.

With future enhancements such as mobile application development, multilingual capabilities, IoT integration, and advanced AI models, Civic Voice has the potential to evolve into a comprehensive smart governance platform. The system contributes to the vision of intelligent, citizen-centric urban management and highlights the role of artificial intelligence in transforming public service delivery.

REFERENCES

- [1] S. S. Mane, A. R. Kulkarni, S. P. Shinde, and D. A. Gavade, "Automatic Complaint Management for Public Services in Smart City," *JETIR*, vol. 6, no. 3, pp. 403–406, 2019.

- [2] S. K. Swarnkar and Y. K. Rathore, "AI-Based Solution to Enable Ease of Grievance Lodging and Tracking," in *Proc. ICAAAI*, Atlantis Press, 2025, pp. 1003–1008.
- [3] D. Banga and K. Peddireddy, "Artificial Intelligence for Customer Complaint Management," *IJCTT*, vol. 71, no. 3, pp. 1–6, 2023.
- [4] S. S. A. Ghadge and R. G. Shinde, "Smart Civic Issue Reporting System," *IJARSC*, vol. 2, no. 1, pp. 252–254, 2022.
- [5] R. T. S. et al., "Smart Maintenance System Using Augmented Reality," in *Proc. IEEE Smart Cities*, 2023, pp. 450–455.
- [6] A. Saha, A. Chaure, A. Bodhe, and T. Bhagat, "Automated Complaint Classification Using NLP and ML," *IJIRT*, vol. 11, no. 6, pp. 756–759, 2024.
- [7] W. G. B. da Silva et al., "Proactive Complaint Management Using AI," *Applied Sciences*, vol. 15, no. 12, p. 6673, 2025.
- [8] Y. Chen and Z. Liu, "Automated Classification of Public Transport Complaints Using LLMs," *Journal of Sensor and Actuator Networks*, vol. 16, no. 8, p. 644, 2025.
- [9] R. K. B. N. et al., "Civic-Fix: Smart Complaint Routing for Urban Solutions," *IJARCC*, vol. 14, no. 6, pp. 197–201, 2025.
- [10] S. Talaat, "Hybrid Deep Learning Models for Text Classification," *Journal of Computer Science*, vol. 21, no. 9, pp. 1983–1992, 2025.
- [11] S. B. R. S. et al., "A Hybrid Text-Image Classification Model for Multimodal Data," *PMC*, vol. 1, pp. 1–10, 2025.
- [12] N. Shukla, "SCMSC: Smart Civic Complaint Management System for Cities," unpublished, 2025.
- [13] G. M. S. C. Smitha, "Enhancing UX in Full Stack Applications Using AI ML," *The Academic Journal*, vol. 3, no. 7, pp. 1106–1110, 2025.
- [14] World Bank Group, "Amplifying Citizens' Voice Through AI-Powered Grievance Systems," World Bank Blog, Aug. 2025.